Linear Circuits One (EEL 3004C)

Practice Exam 2 Solutions

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Practice Exam 2 Q1 Solution

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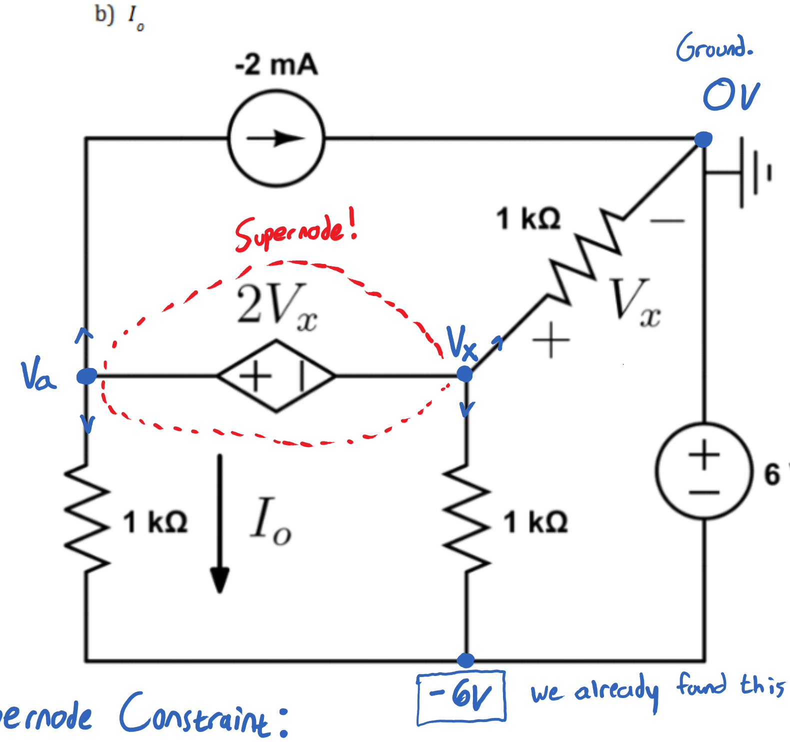
For the given circuit, use nodal analysis to find the following:

a) Every single node voltage

b) I_o



we already found this node voltage!

★ V_x is the voltage difference between its node (V_x) and ground.

★ We don't need to do a KCL at the -6V node because its voltage is given by the source (it is 0-6v, so -6v, due to flipped polarity).

Supernode Constraint:

$$V_a - V_x = 2V_x$$

Simplify:

$$V_a - 3V_x = 0$$

Matrix:

$$\begin{bmatrix} 1 & -3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} V_a \\ V_x \end{bmatrix} = \begin{bmatrix} 0 \\ -10 \end{bmatrix}$$

So,

$$\boxed{V_a = -6V.}$$

$$\boxed{V_x = -2V.}$$

$$i_o = \frac{V_a + 6}{1000}.$$

$$i_o = \frac{-6+6}{1000} = \frac{0}{1000} = 0A.$$

KCL @ V_a and V_x supernode:

$$-0.002 + \frac{V_a + 6}{1000} + \frac{V_x}{1000} + \frac{V_x + 6}{1000} = 0.$$

this is i_o!

Simplify:

$$1000\left(-0.002 + \frac{V_a+6}{1000} + \frac{V_x}{1000} + \frac{V_x+6}{1000} = 0.\right)$$

$$-2 + V_a + 6 + V_x + V_x + 6 = 0.$$

$$-2 + V_a + 6 + 2V_x + 6 = 0$$

$$V_a + 2V_x + 10 = 0$$

$$V_a + 2V_x = -10.$$

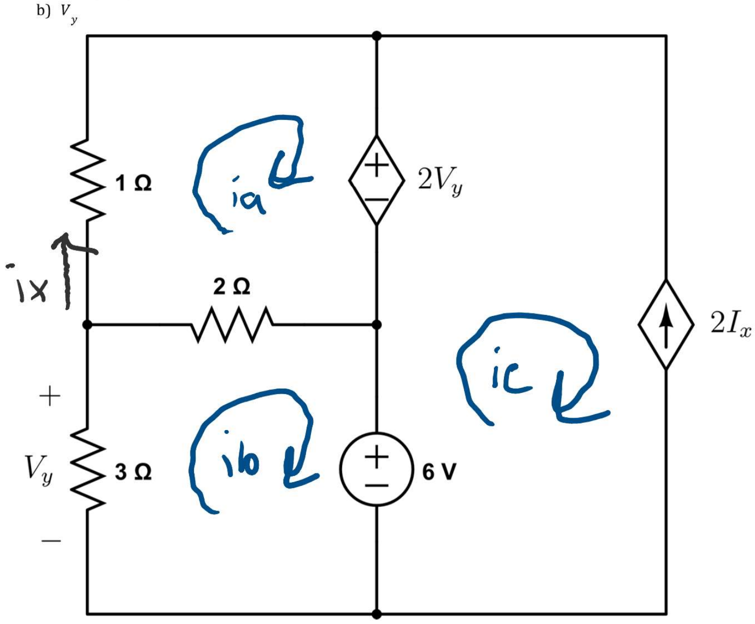
So,

$$\boxed{i_o = 0A.}$$

For the given circuit, use mesh analysis to find the following:

a) Every single mesh current (assume clockwise direction)

b) V_y



5 unknowns, 5 equations

1. $i_x = i_a$

2. $V_y = -3i_b$, $2V_y = -6i_b$

3. $i_c = -2I_x = -2i_a$

Mesh ib: $3i_b + 2(i_b - i_a) + 6 = 0$

Simplify ↳ 4. $-2i_a + 5i_b = -6$

Mesh ia: $i_a + 2V_y + 2(i_a - i_b) = 0$

↳ $i_a - 6i_b + 2i_a - 2i_b = 0$

↳ 5. $3i_a - 8i_b = 0$

Solve for ia, ib:

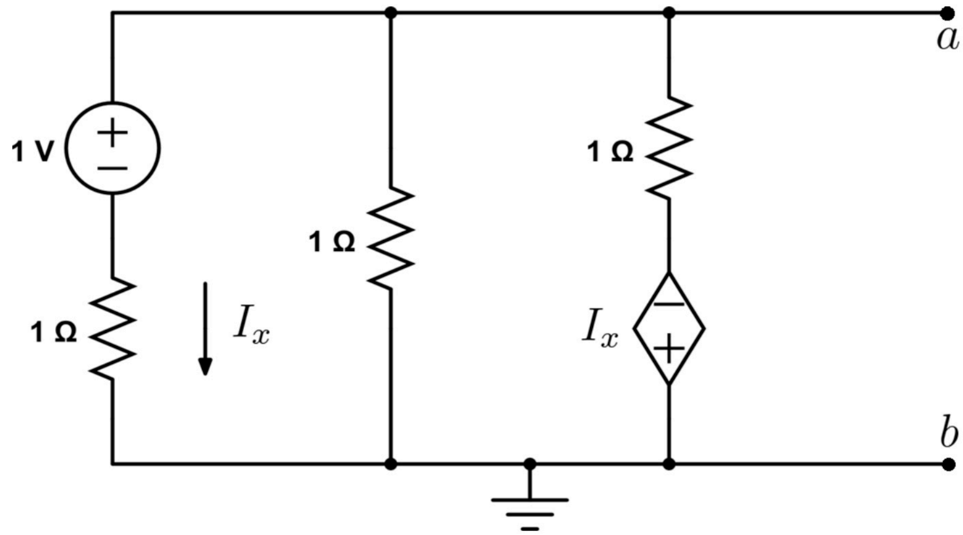
$$\begin{bmatrix} -2 & 5 \\ 3 & -8 \end{bmatrix}^{-1} \cdot \begin{bmatrix} -6 \\ 0 \end{bmatrix} = \begin{bmatrix} i_a \\ i_b \end{bmatrix} = \begin{matrix} 48A \\ 18A \end{matrix}$$

3. $i_c = -2i_a = -96A$

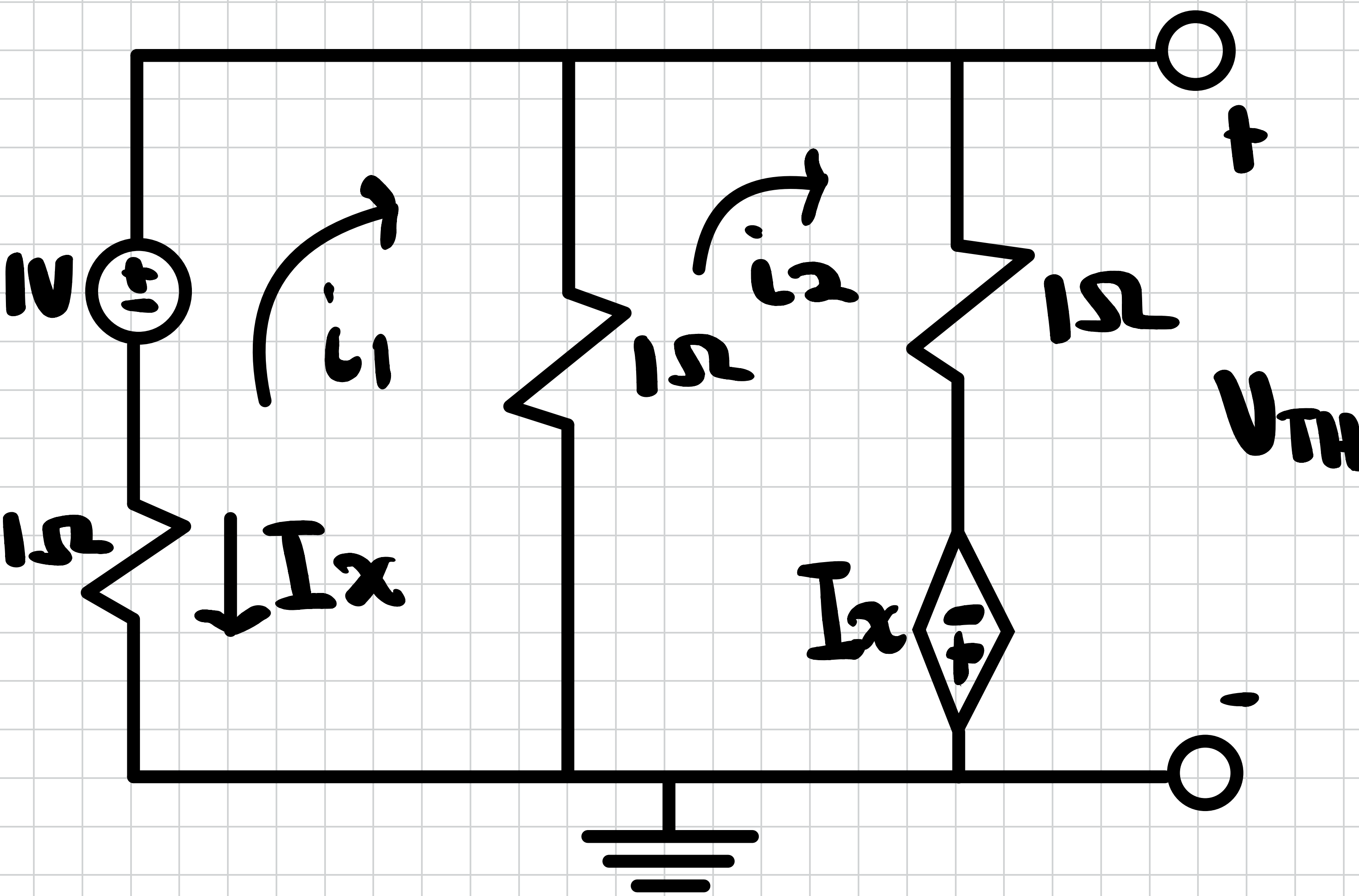
2. $V_y = -3i_b = -54V$

For the given circuit, find the following:

a) The Thevenin equivalent circuit seen from a to b
b) The Norton equivalent circuit seen from a to b
c) The maximum power that could be dissipated in a load resistor R_L connected between a and b



Voc Method



Constraint

$* \; i_1 = -I_x$

Mesh @ i_1

$1(i_1) - 1 + 1(i_1 - i_2) = 0$

$i_1 - 1 + i_1 - i_2 = 0$

$2i_1 - i_2 = 1$

Sub In Constraint

$2(-I_x) - i_2 = 1$

$-2I_x - i_2 = 1 \longrightarrow$ Eq # 1

Mesh @ i_2

$1(i_2 - i_1) + 1(i_2) - I_x = 0$

$i_2 - i_1 + i_2 - I_x = 0$

$-i_1 + 2i_2 - I_x = 0$

Sub In Constraint

$-(-I_x) + i_2 - I_x = 0$

$I_x + i_2 - I_x = 0$

$i_2 = 0A \longrightarrow$ Eq #2

Solve with Both Eq

$-2I_x - 0 = 1$

$I_x = -0.5A$

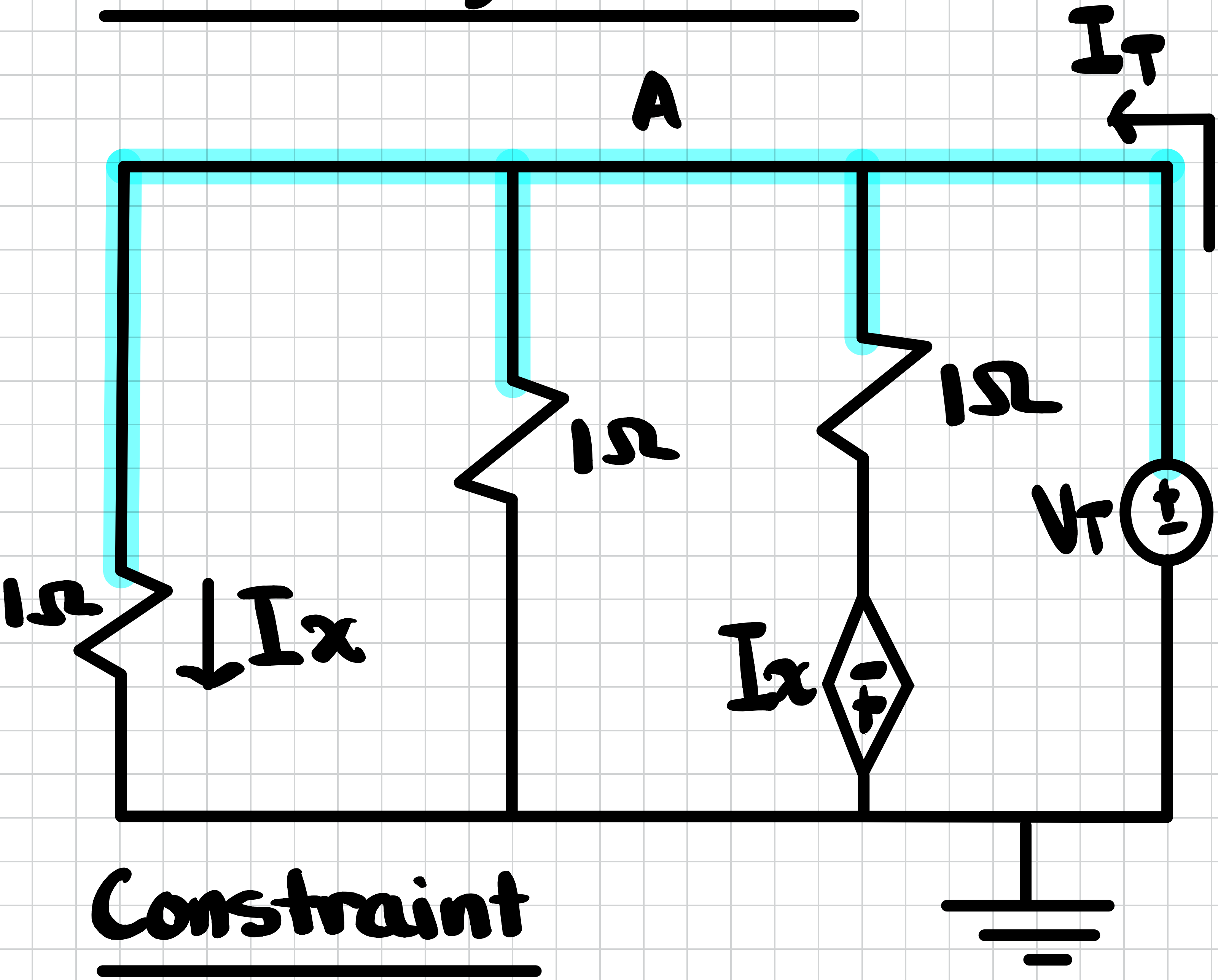
KVL @ Rightmost loop

$I_x + 1i_2 + V_{TH} = 0$

$-0.5 - 1(0) + V_{TH} = 0$

$V_{TH} = 0.5V$

Test Voltage Method



Constraint

* $I_x = \dfrac{V_T - 0}{1} = V_T$

Nodal @ Node A

$$\frac{V_T - 0}{1} + \frac{V_T - 0}{1} + \frac{V_T - (-I_x)}{1} - I_T = 0$$

$$V_T + V_T + V_T + I_x - I_T = 0$$

$$3V_T + I_x - I_T = 0$$

Sub In Constraint

$$3V_T + V_T - I_T = 0$$

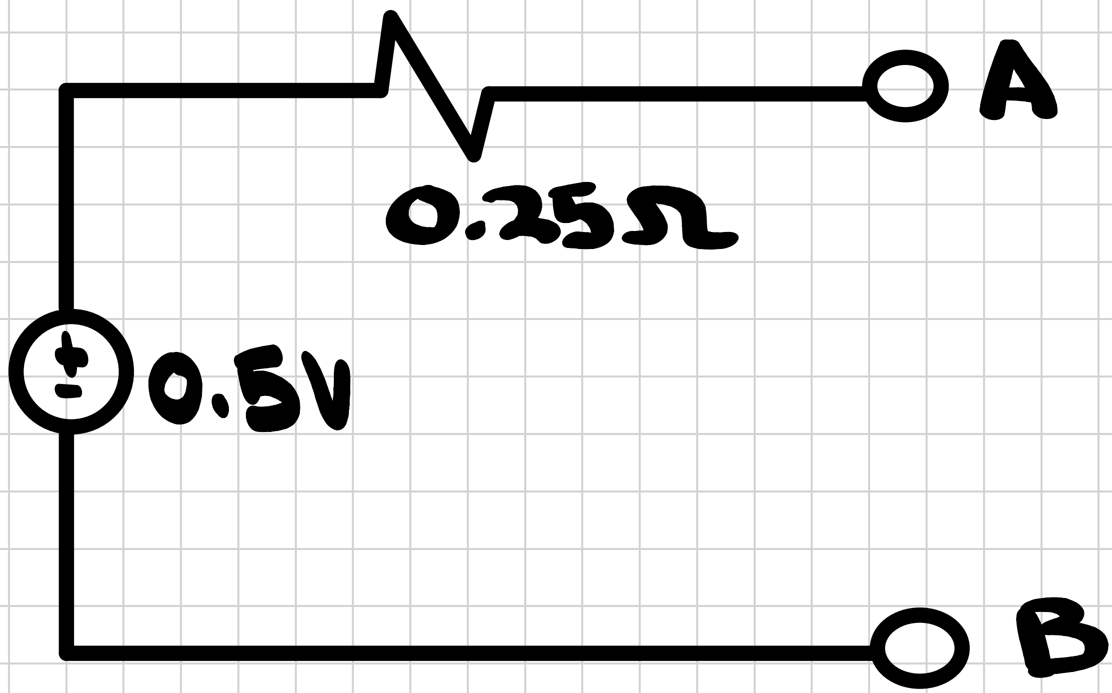
$$4V_T = I_T$$

$$\frac{V_T}{I_T} = 0.25\,\Omega = \frac{1}{4}\,\Omega = R_{TH}$$

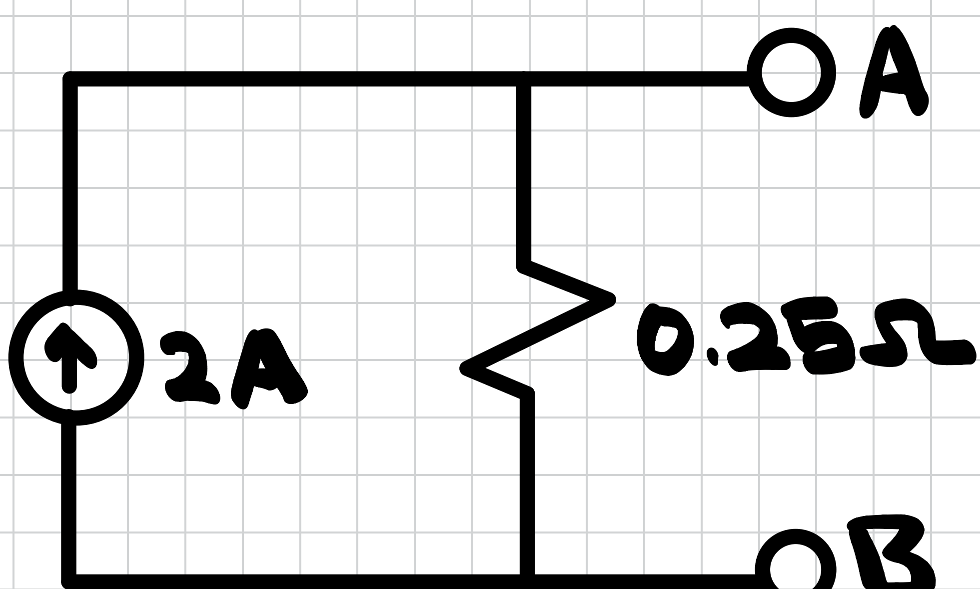
$$I_{Norton} = \frac{V_{TH}}{R_{TH}}$$

$$I_{Norton} = \frac{0.5}{0.25} = 2\,A$$

Thevenin Equivalent



Norton Equivalent



Max Power Transfer

$$P_{MAX} = \frac{V_{TH}^2}{4R_{TH}} = \frac{(0.5)^2}{4(0.25)}$$

$$\boxed{P_{MAX} = 0.25\,W}$$